

PreACT 8/9 Printable Practice Exam 10

Math • Exam 03

35 questions • 40 minutes

PART A - QUESTIONS

M3-001

What is the value of $3^2 - |-5|$?

- A) -14
- B) -4
- C) 4
- D) 14
- E) 86

M3-002

What value of x satisfies $6x - 5 = 37$?

- A) 5
- B) 6
- C) 7
- D) 8
- E) 9

M3-003

An isosceles triangle has a vertex angle measuring 44° . What is the measure of each of the two congruent base angles?

- A) 44°
- B) 56°
- C) 68°
- D) 72°
- E) 136°

M3-004

A candle is 30 centimeters tall when it is lit. It burns at a constant rate, as shown in the table.

t (hours)	0	2	5	8
h (centimeters)	30	26	20	14

Which equation gives the candle's height h after t hours?

- A) $h = 30 - 2t$
- B) $h = 30 - 4t$
- C) $h = 30 + 2t$
- D) $h = 15 - 2t$
- E) $h = 2t$

M3-005

What is the mean of the data set 12, 14, 15, 15, 19?

- A) 14
- B) 15
- C) 15.5

- D) 16
- E) 19

M3-006

What is $(8.4 \times 10^6) + (6 \times 10^5)$ written in scientific notation?

- A) 1.44×10^6
- B) 8.46×10^6
- C) 9×10^6
- D) 9×10^7
- E) 1.44×10^8

M3-007

A store increases the price of a \$50 item by 20%. A customer then uses a coupon for 10% off the increased price. What price does the customer pay before tax?

- A) \$50
- B) \$53
- C) \$54
- D) \$55
- E) \$60

M3-008

A paint mixture uses blue and white paint in the ratio 3:5. If 32 cups of the mixture are made, how many cups are blue paint?

- A) 8 cups
- B) 12 cups
- C) 15 cups
- D) 20 cups
- E) 24 cups

M3-009

A vertical pole is supported by a wire attached to the top of the pole and anchored to the ground. The pole is 9 meters tall, and the anchor is 12 meters from the base of the pole. How long is the wire?

- A) 13 meters
- B) 15 meters
- C) 18 meters
- D) 21 meters
- E) 108 meters

M3-010

A fair number cube has faces numbered 1 through 6. What is the probability of rolling a number greater than 2?

- A) $\frac{1}{3}$
- B) $\frac{1}{2}$
- C) $\frac{2}{3}$
- D) $\frac{3}{4}$
- E) $\frac{5}{6}$

M3-011

What is the value of $\sqrt{196} \div 2 + \sqrt[3]{27}$?

- A) 7

- B) 8
- C) 10
- D) 12
- E) 17

M3-012

The first three terms of an arithmetic sequence are 12, 17, and 22. What is the tenth term?

- A) 45
- B) 52
- C) 55
- D) 57
- E) 62

M3-013

A bus can hold at most 48 people. Five chaperones will ride the bus, and each student group contains 4 students. What is the greatest number of complete student groups that can ride with the chaperones?

- A) 9
- B) 10
- C) 11
- D) 12
- E) 43

M3-014

Point S has coordinates $(-2, 5)$. It is reflected across the line $y = x$. What are the coordinates of the image of S?

- A) $(-5, 2)$
- B) $(-2, -5)$
- C) $(2, -5)$
- D) $(5, -2)$
- E) $(5, 2)$

M3-015

The table shows whether 100 students read fiction regularly.

	Reads fiction	Does not read fiction	Total
Grade 8	28	22	50
Grade 9	36	14	50
Total	64	36	100

One student is selected at random from the students who read fiction regularly. What is the probability that the selected student is in Grade 8?

- A) $\frac{7}{25}$
- B) $\frac{9}{25}$
- C) $\frac{7}{16}$
- D) $\frac{9}{16}$
- E) $\frac{16}{25}$

M3-016

A cyclist travels 24 kilometers in 80 minutes at a constant speed. What is the cyclist's speed in meters per second?

- A) 0.3 meters per second
- B) 3 meters per second

- C) 5 meters per second
- D) 18 meters per second
- E) 300 meters per second

M3-017

A jar contains 26 coins consisting only of nickels and dimes. The coins have a total value of \$2.05. How many dimes are in the jar?

- A) 9
- B) 11
- C) 13
- D) 15
- E) 17

M3-018

An account begins with \$200 and grows by 5% each year. No money is added or withdrawn. What is the account balance after 2 years?

- A) \$210.00
- B) \$215.00
- C) \$220.00
- D) \$220.50
- E) \$225.00

M3-019

A closed cylinder has a radius of 3 centimeters and a height of 7 centimeters. What is the total surface area of the cylinder?

- A) 21π square centimeters
- B) 42π square centimeters
- C) 51π square centimeters
- D) 60π square centimeters
- E) 63π square centimeters

M3-020

The table shows the number of minutes 5 students practiced a skill and the number of errors each student made afterward.

Practice time (minutes)	10	20	30	40	50
Number of errors	18	15	11	8	5

Which statement best describes the association between practice time and number of errors?

- A) There is a positive association because errors generally increase as practice time increases.
- B) There is a negative association because errors generally decrease as practice time increases.
- C) There is no association because the practice times are different.
- D) There is no association because the number of errors is not constant.
- E) There is a positive association because all values are positive.

M3-021

What is the value of $(3^4 \times 3^2) \div 3^3$?

- A) 3
- B) 9
- C) 18
- D) 27
- E) 729

M3-022

The equation $x^2 + 2x - 15 = 0$ has one positive solution. What is that solution?

- A) -5
- B) -3
- C) 3
- D) 5
- E) 15

M3-023

A delivery service charges \$5.20 plus \$1.80 for each mile traveled. If one delivery costs \$19.60, how many miles were traveled?

- A) 6 miles
- B) 7 miles
- C) 8 miles
- D) 9 miles
- E) 11 miles

M3-024

Two rectangles are similar. The smaller rectangle measures 8 centimeters by 12 centimeters. The 12-centimeter side corresponds to an 18-centimeter side of the larger rectangle. What is the area of the larger rectangle?

- A) 144 square centimeters
- B) 180 square centimeters
- C) 192 square centimeters
- D) 216 square centimeters
- E) 324 square centimeters

M3-025

Plan A's savings increase at a constant rate, as shown in the table. Plan B's savings after w weeks are given by $B = 8w + 10$ dollars.

w (weeks)	0	3	6
Plan A savings (dollars)	25	43	61

After 9 weeks, how do the savings under the two plans compare?

- A) Plan A has \$3 more than Plan B.
- B) Plan B has \$3 more than Plan A.
- C) Plan A has \$7 more than Plan B.
- D) Plan B has \$7 more than Plan A.
- E) The two plans have the same amount.

M3-026

A square garden has side length 10 meters. Each side length is increased by 20%. By how many square meters does the garden's area increase?

- A) 20 square meters
- B) 24 square meters
- C) 40 square meters
- D) 44 square meters
- E) 144 square meters

M3-027

What value of x satisfies $(x - 3)/4 = (x + 5)/6$?

- A) 4
- B) 9
- C) 11
- D) 16
- E) 19

M3-028

Which of the following numbers is rational?

- A) $\sqrt{2}$
- B) $\sqrt{7}$
- C) $\sqrt{30}$
- D) $\sqrt{81}$
- E) $\sqrt[3]{10}$

M3-029

A triangle in the coordinate plane has vertices $(0, 0)$, $(6, 0)$, and $(6, 4)$. What is the area of the triangle?

- A) 10 square units
- B) 12 square units
- C) 20 square units
- D) 24 square units
- E) 48 square units

M3-030

Functions f and g are defined by $f(x) = 3x - 2$ and $g(x) = x^2$. What is $f(g(-2))$?

- A) -14
- B) -10
- C) 4
- D) 10
- E) 14

M3-031

A box contains 3 red markers and 2 blue markers. Two markers are selected at random without replacement. What is the probability that both markers are blue?

- A) $1/10$
- B) $1/5$
- C) $1/4$
- D) $2/5$
- E) $1/2$

M3-032

A recipe uses 4.5 cups of flour to make 18 servings. A baker needs enough flour to make 50 servings at the same rate. Flour is sold in bags containing 2.5 cups each. What is the least number of bags needed?

- A) 4
- B) 5
- C) 6
- D) 10

E) 13

M3-033

A rectangle has a perimeter of 54 centimeters. Its length is 3 centimeters more than twice its width. What is the width of the rectangle?

- A) 6 centimeters
- B) 8 centimeters
- C) 9 centimeters
- D) 12 centimeters
- E) 19 centimeters

M3-034

A closed rectangular storage box is 10 centimeters long, 6 centimeters wide, and 4 centimeters high. Every outside face except the bottom is painted. What total area is painted?

- A) 128 square centimeters
- B) 152 square centimeters
- C) 160 square centimeters
- D) 188 square centimeters
- E) 248 square centimeters

M3-035

One class of 12 students has a mean test score of 72. Another class of 8 students has a mean test score of 84. What is the mean test score of all 20 students combined?

- A) 76.0
- B) 76.8
- C) 77.5
- D) 78.0
- E) 78.4

PART B - ANSWER KEY + EXPLANATIONS

M3-001 - Correct: C) 4

Explanation: $3^2 = 9$ and $|-5| = 5$, so $3^2 - |-5| = 9 - 5 = 4$.

M3-002 - Correct: C) 7

Explanation: Add 5 to both sides to get $6x = 42$, and then divide by 6 to get $x = 7$.

M3-003 - Correct: C) 68°

Explanation: The two base angles are congruent and together measure $180^\circ - 44^\circ = 136^\circ$. Each base angle is $136^\circ \div 2 = 68^\circ$.

M3-004 - Correct: A) $h = 30 - 2t$

Explanation: The candle begins at 30 centimeters and loses 4 centimeters every 2 hours, so its rate is -2 centimeters per hour. Therefore, $h = 30 - 2t$.

M3-005 - Correct: B) 15

Explanation: The sum of the 5 values is 75, so the mean is $75 \div 5 = 15$.

M3-006 - Correct: C) 9×10^6

Explanation: Rewrite 6×10^5 as 0.6×10^6 . Then $8.4 \times 10^6 + 0.6 \times 10^6 = 9 \times 10^6$.

M3-007 - Correct: C) \$54

Explanation: A 20% increase changes the price to $\$50 \times 1.20 = \60 . The coupon reduces that price to $\$60 \times 0.90 = \54 .

M3-008 - Correct: B) 12 cups

Explanation: The ratio has $3 + 5 = 8$ total parts. Each part is $32 \div 8 = 4$ cups, so the blue paint amount is $3 \times 4 = 12$ cups.

M3-009 - Correct: B) 15 meters

Explanation: The pole, ground, and wire form a right triangle. The wire length is $\sqrt{9^2 + 12^2} = \sqrt{225} = 15$ meters.

M3-010 - Correct: C) $2/3$

Explanation: Four outcomes, 3, 4, 5, and 6, are greater than 2. The probability is $4/6 = 2/3$.

M3-011 - Correct: C) 10

Explanation: $\sqrt{196} = 14$ and $\sqrt[3]{27} = 3$. Therefore, $14 \div 2 + 3 = 7 + 3 = 10$.

M3-012 - Correct: D) 57

Explanation: The common difference is 5. From the first term to the tenth term there are 9 increases, so the tenth term is $12 + 9 \times 5 = 57$.

M3-013 - Correct: B) 10

Explanation: After seating 5 chaperones, 43 seats remain. Since each group has 4 students, $43 \div 4 = 10$ remainder 3, so at most 10 complete groups can ride.

M3-014 - Correct: D) (5, -2)

Explanation: Reflection across $y = x$ exchanges the coordinates, so $(-2, 5)$ maps to $(5, -2)$.

M3-015 - Correct: C) $7/16$

Explanation: Of the 64 students who read fiction regularly, 28 are in Grade 8. The probability is $28/64 = 7/16$.

M3-016 - Correct: C) 5 meters per second

Explanation: Convert 24 kilometers to 24,000 meters and 80 minutes to 4,800 seconds. The speed is $24,000 \div 4,800 = 5$ meters per second.

M3-017 - Correct: D) 15

Explanation: Let n be nickels and d be dimes. Then $n + d = 26$ and $5n + 10d = 205$. Substituting $n = 26 - d$ gives $130 + 5d = 205$, so $d = 15$.

M3-018 - Correct: D) \$220.50

Explanation: Apply the 5% growth factor twice: $\$200 \times 1.05^2 = \$200 \times 1.1025 = \$220.50$.

M3-019 - Correct: D) 60π square centimeters

Explanation: The total surface area is $2\pi r^2 + 2\pi rh$. With $r = 3$ and $h = 7$, this is $2\pi(3^2) + 2\pi(3)(7) = 18\pi + 42\pi = 60\pi$ square centimeters.

M3-020 - Correct: B) There is a negative association because errors generally decrease as practice time increases.

Explanation: As practice time increases from 10 to 50 minutes, the number of errors decreases from 18 to 5, indicating a negative association.

M3-021 - Correct: D) 27

Explanation: Combine the powers with base 3: $(3^4 \times 3^2) \div 3^3 = 3^3 = 27$.

M3-022 - Correct: C) 3

Explanation: Factoring gives $x^2 + 2x - 15 = (x + 5)(x - 3)$, so the solutions are -5 and 3 . The positive solution is 3 .

M3-023 - Correct: C) 8 miles

Explanation: Subtract the fixed charge: $\$19.60 - \$5.20 = \$14.40$. Then $\$14.40 \div \$1.80 = 8$ miles.

M3-024 - Correct: D) 216 square centimeters

Explanation: The scale factor is $18/12 = 1.5$, so the other side is $8 \times 1.5 = 12$ centimeters. The larger rectangle's area is $12 \times 18 = 216$ square centimeters.

M3-025 - Correct: B) Plan B has \$3 more than Plan A.

Explanation: Plan A increases by \$18 every 3 weeks, or \$6 per week, so after 9 weeks it has $\$25 + 9(\$6) = \$79$. Plan B has $8(9) + 10 = \$82$, which is \$3 more.

M3-026 - Correct: D) 44 square meters

Explanation: The original area is $10^2 = 100$ square meters. Increasing each side by 20% gives side length 12 meters and area $12^2 = 144$ square meters. The increase is $144 - 100 = 44$ square meters.

M3-027 - Correct: E) 19

Explanation: Cross-multiplying gives $6(x - 3) = 4(x + 5)$. Thus, $6x - 18 = 4x + 20$, so $2x = 38$ and $x = 19$.

M3-028 - Correct: D) $\sqrt{81}$

Explanation: $\sqrt{81} = 9$, which is an integer and therefore rational. The other choices are roots of numbers that are not perfect squares or perfect cubes, so they are irrational.

M3-029 - Correct: B) 12 square units

Explanation: The triangle is right with base 6 and height 4. Its area is $(1/2) \times 6 \times 4 = 12$ square units.

M3-030 - Correct: D) 10

Explanation: First, $g(-2) = (-2)^2 = 4$. Then $f(4) = 3(4) - 2 = 10$.

M3-031 - Correct: A) $1/10$

Explanation: The probability of selecting a blue marker first is $2/5$. Without replacement, the probability of another blue marker is then $1/4$. The product is $(2/5)(1/4) = 2/20 = 1/10$.

M3-032 - Correct: B) 5

Explanation: The recipe uses $4.5 \div 18 = 0.25$ cup per serving. For 50 servings, $50 \times 0.25 = 12.5$ cups are needed. Since each bag contains 2.5 cups, $12.5 \div 2.5 = 5$ bags are needed.

M3-033 - Correct: B) 8 centimeters

Explanation: Let w be the width, so the length is $2w + 3$. The perimeter equation is $2(w + 2w + 3) = 54$. This gives $6w + 6 = 54$, so $w = 8$ centimeters.

M3-034 - Correct: D) 188 square centimeters

Explanation: The painted top has area $10 \times 6 = 60$. The four painted side faces have total area $2(10 \times 4) + 2(6 \times 4) = 80 + 48 = 128$. The total painted area is $60 + 128 = 188$ square centimeters.

M3-035 - Correct: B) 76.8

Explanation: The first class has $12 \times 72 = 864$ total score points, and the second has $8 \times 84 = 672$. The combined mean is $(864 + 672) \div 20 = 1,536 \div 20 = 76.8$.